Chapter 19

Covering Groups, Spin Groups, and Spinors

Road map

The preceding chapter exhibited a striking phenomenon:

$$\mathfrak{su}(2) \cong \mathfrak{so}(3), \qquad SU(2)/\{\pm I_2\} \cong SO(3),$$

but $SU(2)$ and $SO(3)$ are not isomorphic as Lie groups. The correct global language is that $SU(2)$ is a *covering group* of $SO(3)$. This chapter develops that idea and then applies it to quantum spin:

1. covering homomorphisms and their discrete central kernels;
2. the universal covering group and the classification of connected covers by subgroups of the fundamental group;
3. the covers of $U(1)$;
4. Pauli spinors and the sign acquired by a spin-$\frac{1}{2}$ state under a 2π rotation;
5. the groups $\mathrm{Spin}(n)$, their low-dimensional accidental isomorphisms, and their centres;
6. the universal and metaplectic covers of $\mathrm{Sp}(2n, \mathbb{R})$;
7. the spin cover $SL(2, \mathbb{C}) \to SO^+(1, 3)$.

19.1 Covering homomorphisms

Suppose G and $\widetilde{G}$ are Lie groups and

$$\Phi : \widetilde{G} \longrightarrow G$$

is a Lie group homomorphism. Its induced Lie-algebra homomorphism is

$$\phi : \widetilde{\mathfrak{g}} \longrightarrow \mathfrak{g}, \qquad \phi(X) = \frac{\mathrm{d}}{\mathrm{d}t}\Phi(\mathrm{e}^{tX})\bigg|_{t=0}.$$

The exponential compatibility diagram is

$$\begin{tikzcd} \widetilde{G} \arrow[r, "\Phi"] & G \\ \widetilde{\mathfrak{g}} \arrow[u, "\exp"] \arrow[r, "\phi"'] & \mathfrak{g} \arrow[u, "\exp"'] \end{tikzcd} \qquad \Phi(\mathrm{e}^{X}) = \mathrm{e}^{\phi(X)}.$$

Definition 19.1: Covering group and covering homomorphism

The Lie group $\widetilde{G}$ is a *covering group* of G, and $\Phi : \widetilde{G} \to G$ is a *covering homomorphism*, if:

1. Φ is surjective;
2. the induced map $\phi : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ is an isomorphism;
3. $\ker \Phi$ is discrete in the ordinary manifold topology.

The three conditions imply the usual topological covering property. Indeed, because $\phi = \mathrm{d}\Phi_{\widetilde{e}}$ is invertible, the inverse function theorem gives an identity neighbourhood $U \subset \widetilde{G}$ on which Φ is a diffeomorphism onto an identity neighbourhood $V \subset G$. Since $\ker \Phi$ is discrete, U may be chosen so that the translates

$$kU, \qquad k \in \ker \Phi,$$

are pairwise disjoint. Every preimage of V differs from its unique representative in U by a kernel element, so

$$\Phi^{-1}(V) = \bigsqcup_{k \in \ker \Phi} kU.$$

Each kU maps diffeomorphically onto V. Translating this picture over every point of G proves that Φ is a covering map.

The quotient description

Every covering homomorphism satisfies

$$G \cong \widetilde{G}/\ker \Phi.$$

The Lie algebra does not see the kernel because a discrete subgroup has zero-dimensional tangent space.

19.1.1 A discrete kernel is central when the cover is connected

Proposition 19.1: Centrality of the kernel

Let $\Phi : \widetilde{G} \to G$ be a covering homomorphism. If $\widetilde{G}$ is connected, then

$$\ker \Phi \subseteq Z(\widetilde{G}). \tag{19.1}$$

In particular, $\ker \Phi$ is abelian.

证明. Take $k \in \ker\Phi$ and $g \in \widetilde{G}$. Because $\widetilde{G}$ is connected, it is path-connected, so there is a smooth path

$$g : [0,1] \longrightarrow \widetilde{G}, \qquad g(0) = \widetilde{e}, \quad g(1) = g.$$

Define

$$f(t) = g(t)kg(t)^{-1}.$$

Then

$$\begin{aligned}\Phi(f(t)) &= \Phi(g(t))\Phi(k)\Phi(g(t))^{-1}\\ &= \Phi(g(t))e\Phi(g(t))^{-1} = e.\end{aligned}$$

Thus $f(t) \in \ker\Phi$ for every t. The map f is continuous, whereas $\ker\Phi$ is discrete. Since $[0,1]$ is connected, f must be constant. Therefore

$$f(0) = k = f(1) = gkg^{-1}.$$

As g was arbitrary, $k \in Z(\widetilde{G})$. ■

This argument is often summarized by saying that a connected covering group "wraps around" the base group, and each deck displacement lies in its centre.

19.2 First examples

19.2.1 The double cover of $SO(3)$

The map constructed in the preceding chapter,

$$R : SU(2) \longrightarrow SO(3),$$

has all the required properties:

$$R \text{ is surjective}, \qquad \mathrm{d}R_e : \mathfrak{su}(2) \xrightarrow{\sim} \mathfrak{so}(3), \qquad \ker R = \{\pm I_2\} \cong \mathbb{Z}_2.$$

Hence $SU(2)$ is a double cover of $SO(3)$.

19.2.2 The exponential cover of the circle

Let

$$\Phi : \mathbb{R} \longrightarrow U(1), \qquad t \longmapsto \mathrm{e}^{\mathrm{i}t}. \tag{19.2}$$

This is a continuous group homomorphism because

$$\Phi(t_1 + t_2) = \mathrm{e}^{\mathrm{i}(t_1+t_2)} = \mathrm{e}^{\mathrm{i}t_1}\mathrm{e}^{\mathrm{i}t_2} = \Phi(t_1)\Phi(t_2).$$

It is surjective. Its Lie-algebra map is

$$\mathbb{R} \longrightarrow \mathrm{i}\mathbb{R}, \qquad x \longmapsto \mathrm{i}x,$$

which is an isomorphism of one-dimensional abelian real Lie algebras. Its kernel is

$$\ker \Phi = \{t \in \mathbb{R} : \mathrm{e}^{\mathrm{i}t} = 1\} = 2\pi\mathbb{Z}. \tag{19.3}$$

Thus

$$\mathbb{R}/(2\pi\mathbb{Z}) \cong U(1).$$

The covering group $\mathbb{R}$ is simply connected, while the base circle is not.

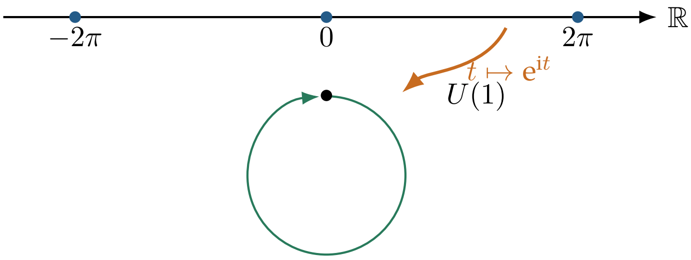

图 **19.1:** Each interval of length 2π wraps once around the circle.

There are also non-universal connected covers of the circle. For a fixed integer $n \geq 1$,

$$\Phi_n : U(1) \longrightarrow U(1), \qquad z \longmapsto z^n, \tag{19.4}$$

is an n-sheeted covering homomorphism. Its kernel is the group of n-th roots of unity,

$$\ker \Phi_n = \left\{\mathrm{e}^{2\pi \mathrm{i}k/n} : k = 0, 1, \ldots, n-1\right\} \cong \mathbb{Z}_n.$$

The covering group is again $U(1)$, and so it is not simply connected. This example shows that a cover of a Lie group is not unique.

19.3 The universal covering group

Definition 19.2: Universal cover

Let G be a connected Lie group with Lie algebra $\mathfrak{g}$. A *universal covering group* of G is a connected, simply connected Lie group $\widetilde{G}_{\text{univ}}$ with

$$\mathrm{Lie}(\widetilde{G}_{\text{univ}}) \cong \mathfrak{g}$$

and a surjective covering homomorphism

$$p : \widetilde{G}_{\text{univ}} \longrightarrow G.$$

Theorem 19.1: Existence and uniqueness of the universal group cover

Every connected Lie group has a universal covering group, unique up to a Lie group isomorphism commuting with the covering maps. For its covering homomorphism p,

$$\ker p \cong \pi_1(G) \subseteq Z(\widetilde{G}_{\text{univ}}), \qquad G \cong \widetilde{G}_{\text{univ}} / \ker p. \tag{19.5}$$

The universal cover “unwraps” every closed path that cannot be deformed to the identity loop while keeping its base point fixed. If G is disconnected, one first applies the construction to the identity component G°; the Lie algebra controls only that component.

19.3.1 Every connected cover comes from the universal one

Let
$$p : \widetilde{G}_{\text{univ}} \longrightarrow G, \qquad K = \ker p \cong \pi_1(G).$$
For every subgroup $H \leq K$, the quotient
$$\widetilde{G}_{\text{univ}}/H$$
is a connected covering group of G. Conversely, every connected covering group of G is isomorphic to one of these quotients. The extreme cases are
$$H = \{e\} : \widetilde{G}_{\text{univ}}, \qquad H = K : G.$$
Thus the universal cover is maximal: all other connected covers are its quotients by subgroups of the discrete central kernel.

For the circle,
$$\widetilde{U(1)}_{\text{univ}} = \mathbb{R}, \qquad K = 2\pi\mathbb{Z}.$$
Quotienting by $2\pi n\mathbb{Z}$ gives another circle, and the induced map to $U(1)$ is the n-sheeted cover $z \mapsto z^n$.

19.4 Pauli spinors in a magnetic field

Consider a spin-$\frac{1}{2}$ particle in a nonzero external magnetic field $\boldsymbol{B} \in \mathbb{R}^3$. In convenient units its Hamiltonian is
$$H(\boldsymbol{B}) = -\frac{1}{2}\boldsymbol{B} \cdot \boldsymbol{\sigma}. \tag{19.6}$$
The Hilbert space is $\mathbb{C}^2$, and the spin observable is
$$\boldsymbol{S} = \frac{1}{2}\boldsymbol{\sigma}.$$
Because
$$(\boldsymbol{B} \cdot \boldsymbol{\sigma})^2 = \|\boldsymbol{B}\|^2 I_2,$$
the eigenvalues of $\boldsymbol{B} \cdot \boldsymbol{\sigma}$ are $\pm\|\boldsymbol{B}\|$. Therefore the two energy levels are
$$E_0(\boldsymbol{B}) = -\frac{\|\boldsymbol{B}\|}{2}, \qquad E_1(\boldsymbol{B}) = +\frac{\|\boldsymbol{B}\|}{2}.$$
This is the Zeeman splitting.

19.4.1 The direction-dependent ground state

Write

$$\boldsymbol{B} = \|\boldsymbol{B}\| \begin{pmatrix} \cos\varphi \sin\vartheta \\ \sin\varphi \sin\vartheta \\ \cos\vartheta \end{pmatrix} = \|\boldsymbol{B}\|\, \boldsymbol{n}, \qquad \|\boldsymbol{n}\| = 1. \tag{19.7}$$

A normalized ground-state spinor is

$$\boxed{|\psi_0(\vartheta,\varphi)\rangle = \begin{pmatrix} \cos(\vartheta/2) \\ \mathrm{e}^{\mathrm{i}\varphi}\sin(\vartheta/2) \end{pmatrix}.} \tag{19.8}$$

It depends only on the direction of $\boldsymbol{B}$, not on its magnitude.

To check the eigenvalue equation, note that

$$\boldsymbol{n}\cdot\boldsymbol{\sigma} = \begin{pmatrix} \cos\vartheta & \mathrm{e}^{-\mathrm{i}\varphi}\sin\vartheta \\ \mathrm{e}^{\mathrm{i}\varphi}\sin\vartheta & -\cos\vartheta \end{pmatrix}.$$

Multiplication gives

$$\begin{aligned}(\boldsymbol{n}\cdot\boldsymbol{\sigma})|\psi_0\rangle &= \begin{pmatrix} \cos\vartheta\cos(\vartheta/2) + \sin\vartheta\sin(\vartheta/2) \\ \mathrm{e}^{\mathrm{i}\varphi}\big(\sin\vartheta\cos(\vartheta/2) - \cos\vartheta\sin(\vartheta/2)\big) \end{pmatrix} \\ &= \begin{pmatrix} \cos(\vartheta/2) \\ \mathrm{e}^{\mathrm{i}\varphi}\sin(\vartheta/2) \end{pmatrix} = |\psi_0\rangle.\end{aligned}$$

Hence

$$H(\boldsymbol{B})|\psi_0\rangle = -\frac{\|\boldsymbol{B}\|}{2}|\psi_0\rangle.$$

19.4.2 The spin expectation aligns with the field

Put

$$c = \cos(\vartheta/2), \qquad s = \sin(\vartheta/2).$$

Then the three component calculations are

$$\begin{aligned}\langle\psi_0|\sigma_1|\psi_0\rangle &= (c, \mathrm{e}^{-\mathrm{i}\varphi}s)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} c \\ \mathrm{e}^{\mathrm{i}\varphi}s \end{pmatrix} \\ &= cs(\mathrm{e}^{\mathrm{i}\varphi} + \mathrm{e}^{-\mathrm{i}\varphi}) = 2cs\cos\varphi = \sin\vartheta\cos\varphi, \\ \langle\psi_0|\sigma_2|\psi_0\rangle &= (c, \mathrm{e}^{-\mathrm{i}\varphi}s)\begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}\begin{pmatrix} c \\ \mathrm{e}^{\mathrm{i}\varphi}s \end{pmatrix} \\ &= -\mathrm{i}cs\mathrm{e}^{\mathrm{i}\varphi} + \mathrm{i}cs\mathrm{e}^{-\mathrm{i}\varphi} = \sin\vartheta\sin\varphi, \\ \langle\psi_0|\sigma_3|\psi_0\rangle &= (c, \mathrm{e}^{-\mathrm{i}\varphi}s)\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} c \\ \mathrm{e}^{\mathrm{i}\varphi}s \end{pmatrix} \\ &= c^2 - s^2 = \cos\vartheta.\end{aligned}$$

The half-angle identities

$$2\cos(\vartheta/2)\sin(\vartheta/2) = \sin\vartheta, \qquad \cos^2(\vartheta/2) - \sin^2(\vartheta/2) = \cos\vartheta$$

were used in the last steps. Therefore

$$\boxed{\langle\psi_0|\boldsymbol{S}|\psi_0\rangle = \frac{1}{2}\boldsymbol{n} = \frac{1}{2}\frac{\boldsymbol{B}}{\|\boldsymbol{B}\|}.} \tag{19.9}$$

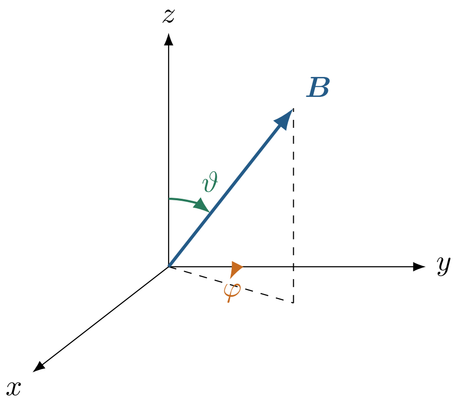

图 **19.2:** The ground-state spin expectation points along the magnetic-field direction.

19.5 Rotations of a spinor

We use the active viewpoint: the physical state is rotated,

$$|\psi\rangle \longmapsto |\psi'\rangle = U(R)|\psi\rangle,$$

while the laboratory measurement apparatus and the operators $\boldsymbol{S} = \frac{1}{2}\boldsymbol{\sigma}$ are kept fixed.

Under an ordinary rotation $R \in SO(3)$, a classical vector transforms as $\boldsymbol{v} \mapsto R\boldsymbol{v}$. Therefore the quantum expectation value must obey

$$\langle\psi'|\boldsymbol{S}|\psi'\rangle = R\langle\psi|\boldsymbol{S}|\psi\rangle.$$

Since this should hold for every state, the corresponding operator identity is

$$\boxed{U(R)^\dagger \boldsymbol{S}\, U(R) = R\boldsymbol{S}.} \tag{19.10}$$

In components,

$$U(R)^\dagger S_\alpha U(R) = \sum_{\beta=1}^{3} R_{\alpha\beta} S_\beta.$$

Wigner's theorem says that a symmetry preserving transition probabilities acts on rays by a unitary or antiunitary operator. Rotations vary continuously from the identity, so the relevant implementation is unitary. Its phase can be chosen locally so that, for an infinitesimal rotation, $U(R)$ is close to I_2; in particular $U(R) \in SU(2)$.

For a rotation

$$R = \mathrm{e}^{\theta T(\boldsymbol{n})} \in SO(3),$$

the double-cover calculation of the preceding chapter gives two possible lifts:

$$U_{\pm}(R) = \pm \exp\left(-\frac{\mathrm{i}\theta}{2}\boldsymbol{n}\cdot\boldsymbol{\sigma}\right). \tag{19.11}$$

The lift continuous from the identity is

$$U(R) = \exp\left(-\frac{\mathrm{i}\theta}{2}\boldsymbol{n}\cdot\boldsymbol{\sigma}\right).$$

Equation (19.10) is exactly the adjoint identity

$$U^\dagger \sigma_\alpha U = \sum_\beta R_{\alpha\beta}\sigma_\beta.$$

19.5.1 A 2π rotation changes the spinor's sign

Because $(\boldsymbol{n}\cdot\boldsymbol{\sigma})^2 = I_2$,

$$\begin{aligned} U(2\pi, \boldsymbol{n}) &= \exp(-\mathrm{i}\pi\, \boldsymbol{n}\cdot\boldsymbol{\sigma}) \\ &= \cos\pi\, I_2 - \mathrm{i}\sin\pi\, \boldsymbol{n}\cdot\boldsymbol{\sigma} = -I_2. \end{aligned} \tag{19.12}$$

Thus a $360°$ rotation acts by

$$|\psi\rangle \longmapsto -|\psi\rangle.$$

A second full turn gives $U(4\pi, \boldsymbol{n}) = I_2$.

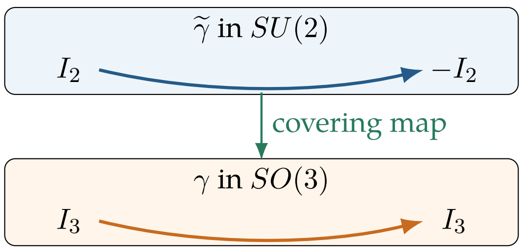

图 **19.3:** The closed 2π-rotation loop in $SO(3)$ lifts to a path from I_2 to $-I_2$ in $SU(2)$.

The sign is a quantum-mechanical phase. A single isolated ray does not distinguish $|\psi\rangle$ from $-|\psi\rangle$, but the relative sign is observable in interference. It is the representation-theoretic shadow of the nontrivial loop in $\pi_1(SO(3)) \cong \mathbb{Z}_2$, and it may be described as a topological or Berry phase of π.

Definition 19.3: Pauli spinor

A two-component complex object transforming according to the defining representation of $SU(2)$,

$$\psi \longmapsto U\psi, \qquad U \in SU(2),$$

is called a *Pauli spinor*. Its representation space is $\mathbb{C}^2$.

19.6 Spin groups

The group $SO(n)$ is connected for every $n \geq 2$. Its fundamental group is

$$\pi_1(SO(2)) \cong \mathbb{Z}, \qquad \pi_1(SO(n)) \cong \mathbb{Z}_2 \quad (n \geq 3). \tag{19.13}$$

For $n \geq 4$, the fibration

$$SO(n-1) \longrightarrow SO(n) \longrightarrow S^{n-1}, \qquad A \longmapsto Ae_n,$$

and its long exact homotopy sequence reduce $\pi_1(SO(n))$ inductively to $\pi_1(SO(3))$. The base sphere S^{n-1} is simply connected for $n \geq 3$, which is why the answer stabilizes at $\mathbb{Z}_2$.

Definition 19.4: Spin group

For $n \geq 3$, the *spin group* is the universal covering group

$$\mathrm{Spin}(n) := \widetilde{SO(n)}_{\mathrm{univ}}.$$

It has Lie algebra

$$\mathrm{Lie}(\mathrm{Spin}(n)) \cong \mathfrak{so}(n).$$

There is a canonical double covering homomorphism

$$\rho_n : \mathrm{Spin}(n) \longrightarrow SO(n), \qquad \ker \rho_n = \{\pm 1\}, \tag{19.14}$$

and hence

$$SO(n) \cong \mathrm{Spin}(n)/\{\pm 1\} \qquad (n \geq 3). \tag{19.15}$$

The exceptional notation in dimension two

The universal cover of $SO(2) \cong U(1)$ is

$$\mathbb{R} \longrightarrow SO(2), \qquad t \longmapsto \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix},$$

with kernel $2\pi\mathbb{Z}$. Standard Clifford-algebra notation instead uses $\mathrm{Spin}(2) \cong U(1)$ for the *double* cover $z \mapsto z^2$ of $SO(2)$. Thus the conventional compact $\mathrm{Spin}(2)$ is not the universal cover. Keeping these two objects distinct resolves the apparent conflict in the low-dimensional table.

19.6.1 Low-dimensional spin groups and their centres

Several low-dimensional spin groups are isomorphic to more familiar classical groups. These are sometimes called accidental isomorphisms:

$$\mathrm{Spin}(3) \cong SU(2), \qquad \mathrm{Spin}(4) \cong SU(2) \times SU(2),$$

$$\mathrm{Spin}(5) \cong Sp(2), \qquad \mathrm{Spin}(6) \cong SU(4).$$

Here $Sp(2)$ means the compact symplectic group of 2×2 quaternionic unitary matrices, equivalently $USp(4)$; it should not be confused with the real group $\mathrm{Sp}(4, \mathbb{R})$.

n	$\dim SO(n)$	$\pi_1(SO(n))$	$\widetilde{SO(n)}_{\text{univ}}$	kernel	centre of the cover	$Z(SO(n))$
2	1	$\mathbb{Z}$	$\mathbb{R}$	$2\pi\mathbb{Z}$	$\mathbb{R}$	$SO(2)$
3	3	$\mathbb{Z}_2$	$SU(2)$	$\{\pm 1\}$	$\mathbb{Z}_2$	$\{1\}$
4	6	$\mathbb{Z}_2$	$SU(2) \times SU(2)$	$\{\pm(1,1)\}$	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2$
5	10	$\mathbb{Z}_2$	$Sp(2)$	$\{\pm 1\}$	$\mathbb{Z}_2$	$\{1\}$
6	15	$\mathbb{Z}_2$	$SU(4)$	$\{\pm 1\}$	$\mathbb{Z}_4$	$\mathbb{Z}_2$
7	21	$\mathbb{Z}_2$	$\mathrm{Spin}(7)$	$\{\pm 1\}$	$\mathbb{Z}_2$	$\{1\}$
8	28	$\mathbb{Z}_2$	$\mathrm{Spin}(8)$	$\{\pm 1\}$	$\mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_2$

表 **19.1:** The low-dimensional covering groups, kernels, and centres.

The dimension formula in the table is

$$\dim SO(n) = \dim \mathfrak{so}(n) = \frac{n(n-1)}{2}.$$

For $n \geq 3$, the centre of $\mathrm{Spin}(n)$ follows the pattern

$$Z(\mathrm{Spin}(n)) \cong \begin{cases} \mathbb{Z}_2, & n \text{ odd}, \\ \mathbb{Z}_4, & n \equiv 2 \pmod 4, \\ \mathbb{Z}_2 \times \mathbb{Z}_2, & n \equiv 0 \pmod 4, \end{cases} \tag{19.16}$$

whereas

$$Z(SO(n)) = \begin{cases} \{I_n\}, & n \text{ odd}, \\ \{\pm I_n\} \cong \mathbb{Z}_2, & n \text{ even} \end{cases} \qquad (n \geq 3). \tag{19.17}$$

The covering kernel $\{\pm 1\}$ is always central, but it need not be the entire centre of $\mathrm{Spin}(n)$.

19.7 Two further families of covers

19.7.1 The special unitary groups

For every $n \geq 2$, $SU(n)$ is simply connected. One way to see the inductive structure is through the fibration

$$SU(n-1) \longrightarrow SU(n) \longrightarrow S^{2n-1}, \qquad U \longmapsto U\boldsymbol{e}_n.$$

The base S^{2n-1} is simply connected, and the induction starts with

$$SU(2) \cong S^3.$$

Also $SU(1) = \{1\}$. Therefore there is nothing further to unwrap:

$$\widetilde{SU(n)}_{\text{univ}} = SU(n).$$

19.7.2 The real symplectic group and the metaplectic group

The real symplectic group

$$\mathrm{Sp}(2n,\mathbb{R}) = \left\{ M \in GL(2n,\mathbb{R}) : M^{\mathsf{T}} J M = J \right\}, \qquad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

is connected but not simply connected. Its fundamental group and centre are

$$\pi_1(\mathrm{Sp}(2n,\mathbb{R})) \cong \mathbb{Z}, \qquad Z(\mathrm{Sp}(2n,\mathbb{R})) = \{\pm I_{2n}\}. \tag{19.18}$$

Consequently its universal covering homomorphism

$$p : \widetilde{\mathrm{Sp}}(2n,\mathbb{R}) \longrightarrow \mathrm{Sp}(2n,\mathbb{R})$$

has infinitely many sheets and

$$\ker p \cong \mathbb{Z}.$$

The universal covering group $\widetilde{\mathrm{Sp}}(2n,\mathbb{R})$ is a genuine Lie group, but it is not a linear matrix Lie group: no faithful finite-dimensional matrix representation realizes it as a closed subgroup of some $GL(N,\mathbb{C})$.

The subgroup $2\mathbb{Z}$ of the universal kernel gives an important intermediate quotient:

$$\mathrm{Mp}(2n,\mathbb{R}) := \widetilde{\mathrm{Sp}}(2n,\mathbb{R})/(2\mathbb{Z}). \tag{19.19}$$

There is a double covering homomorphism

$$\mathrm{Mp}(2n,\mathbb{R}) \longrightarrow \mathrm{Sp}(2n,\mathbb{R}), \qquad \ker = \{\pm 1\}. \tag{19.20}$$

The group $\mathrm{Mp}(2n,\mathbb{R})$ is the *metaplectic group*. The quotient relations are

$$\mathrm{Mp}(2n,\mathbb{R}) \cong \widetilde{\mathrm{Sp}}(2n,\mathbb{R})/(2\mathbb{Z}), \qquad \mathrm{Sp}(2n,\mathbb{R}) \cong \widetilde{\mathrm{Sp}}(2n,\mathbb{R})/\mathbb{Z},$$

and hence

$$\mathrm{Sp}(2n,\mathbb{R}) \cong \mathrm{Mp}(2n,\mathbb{R})/\mathbb{Z}_2.$$

$$\begin{tikzcd} \widetilde{\mathrm{Sp}}(2n,\mathbb{R}) \arrow[r, "\text{kernel } 2\mathbb{Z}"] \arrow[rr, bend left, "\text{kernel } \mathbb{Z}"] & \mathrm{Mp}(2n,\mathbb{R}) \arrow[r, "\text{kernel } \mathbb{Z}_2"] & \mathrm{Sp}(2n,\mathbb{R}) \end{tikzcd}$$

图 **19.4:** The universal and metaplectic covers of the real symplectic group.

This resembles $SO(2)$: both $SO(2)$ and $\mathrm{Sp}(2n,\mathbb{R})$ have fundamental group $\mathbb{Z}$, so both possess an infinite universal cover and nontrivial finite intermediate covers. The essential difference is that $SO(2)$ is abelian and compact, whereas $\mathrm{Sp}(2n,\mathbb{R})$ is noncompact and semisimple.

19.8 The proper orthochronous Lorentz group

The full Lorentz group $SO(1,3)$ is not connected. The component containing the identity is the proper orthochronous Lorentz group

$$SO^+(1,3).$$

Its Lie algebra is

$$\mathfrak{so}(1,3) = \left\{ X \in \mathfrak{gl}(4,\mathbb{R}) : X^\mathsf{T}\eta + \eta X = 0 \right\}, \qquad \eta = \operatorname{diag}(1,-1,-1,-1). \tag{19.21}$$

As real Lie algebras,

$$\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C}).$$

Furthermore,

$$\pi_1(SO^+(1,3)) \cong \mathbb{Z}_2,$$

and its universal covering group is

$$\mathrm{Spin}^+(1,3) \cong SL(2,\mathbb{C}).$$

The covering map can be seen explicitly. Associate

$$x = (x^0, x^1, x^2, x^3) \in \mathbb{R}^{1,3}$$

with the Hermitian matrix

$$X = x^0 I_2 + \sum_{j=1}^{3} x^j \sigma_j.$$

Its determinant is the Minkowski quadratic form:

$$\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2.$$

For $A \in SL(2,\mathbb{C})$, define

$$X \longmapsto AXA^\dagger.$$

This preserves Hermiticity and determinant, so it determines a Lorentz transformation $\Lambda(A) \in SO^+(1,3)$. The assignment

$$\Lambda : SL(2,\mathbb{C}) \longrightarrow SO^+(1,3)$$

is a surjective Lie group homomorphism with

$$\ker \Lambda = \{\pm I_2\}.$$

Therefore

$$\boxed{SO^+(1,3) \cong SL(2,\mathbb{C})/\{\pm I_2\}.} \tag{19.22}$$

This is the Lorentzian analogue of

$$SO(3) \cong SU(2)/\{\pm I_2\}.$$

What to retain from this chapter

1. A covering homomorphism $\Phi : \widetilde{G} \to G$ is surjective, induces an isomorphism of Lie algebras, and has discrete kernel. It realizes
$$G \cong \widetilde{G}/\ker \Phi.$$
2. If $\widetilde{G}$ is connected, then
$$\ker \Phi \subseteq Z(\widetilde{G}).$$
The proof conjugates a kernel element along a path and uses discreteness.
3. The universal cover of a connected Lie group is connected and simply connected, and
$$\ker p \cong \pi_1(G).$$
Every connected cover is obtained by quotienting the universal cover by a subgroup of this kernel.
4. For the circle,
$$\mathbb{R} \to U(1), \quad t \mapsto e^{it}, \qquad \ker = 2\pi\mathbb{Z},$$
while $z \mapsto z^n$ gives non-universal n-sheeted covers.
5. A Pauli spinor transforms by
$$U(R) = \exp\left(-\frac{i\theta}{2}\boldsymbol{n} \cdot \boldsymbol{\sigma}\right),$$
and
$$U(2\pi, \boldsymbol{n}) = -I_2, \qquad U(4\pi, \boldsymbol{n}) = I_2.$$
6. For $n \geq 3$,
$$\mathrm{Spin}(n) = \widetilde{SO(n)}_{\text{univ}}, \qquad SO(n) \cong \mathrm{Spin}(n)/\{\pm 1\}.$$
In particular, $\mathrm{Spin}(3) \cong SU(2)$.
7. The real symplectic group has
$$\pi_1(\mathrm{Sp}(2n, \mathbb{R})) \cong \mathbb{Z}.$$
Its universal cover has infinitely many sheets, while $\mathrm{Mp}(2n, \mathbb{R})$ is its distinguished double cover.
8. The identity component of the Lorentz group satisfies
$$SO^+(1,3) \cong SL(2, \mathbb{C})/\{\pm I_2\}.$$